Prácticas Finales/EPS

Modificaciones al entorno MadGraph

Escuela de Ciencias Físicas y Matemáticas
Universidad de San Carlos de Guatemala

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I. Modulos del detector Delphes

A. TrackCovariance

A partir del método vacío *void TrackCovariance::Process()*

```
void TrackCovariance::Process()
{
  Candidate *candidate, *mother, *particle;
  Double_t mass, p, pt, q, ct;
  Double_t dd0, ddz, dphi, dct, dp, dpt, dC;
  // (...)
  fItInputArray->Reset();
  while((candidate = static_cast<Candidate *>(fItInputArray->Next())))
  {
    // (...)
    dd0        = TMath::Sqrt(track.GetCov()(0, 0))*1e03;
    ddz        = TMath::Sqrt(track.GetCov()(3, 3))*1e03;
    dphi       = TMath::Sqrt(track.GetCov()(1, 1));
    dct        = TMath::Sqrt(track.GetCov()(4, 4));
    dpt        = 2 * TMath::Sqrt( track.GetCov()(2, 2))*pt*pt / (0.2998*fBz);
    dp         = TMath::Sqrt((1.+ct*ct)*dpt*dpt + 4*pt*pt*ct*ct*dct*dct/(1.+ct*ct)/(1.+ct*ct));
    dC         = TMath::Sqrt(track.GetCov()(2, 2))*1e-03;

    candidate->ErrorD0 = dd0;
    candidate->ErrorDZ = ddz;
    candidate->ErrorP = dp;
    candidate->ErrorC = dC;
    candidate->ErrorCtgTheta = dct;
    candidate->ErrorPhi = dphi;
    candidate->ErrorPT = dpt;
    //candidate->TrackResolution = dpt / pt;
    candidate->TrackResolution = dp / p;
    // (...)
  }
}
```

se definen las variables para generar los valores de *D0*, *DZ*, *ErrorD0*, *ErrorDZ*, *ErrorPT*, *ErrorP*, *ErrorPhi* y la matriz completa *TrackCovariance*. Por tanto, con este archivo se deberían calcular las incertidumbres como las raíces de los elementos diagonales de la matriz de covarianza.
Sin embargo, al momento de verificar los valores de las incertidumbres *ErrorD0* y *ErrorDZ* para un proceso

```
# Para los modelos sm como sm-full
MG5_aMC> generate p p > mu+ mu-
```

se obtiene que

```
import UpROOT.io as io
io.load_dataframe(
    branches=('Muon/Muon.D0', 'Muon/Muon.DZ', 'Muon/Muon.ErrorD0', 'Muon/Muon.ErrorDZ'),
    dataset=DATASET,
).head(30)
```

Dando como salida la siguiente tabla

	Muon/Muon.D0	Muon/Muon.DZ	Muon/Muon.ErrorD0	Muon/Muon.ErrorDZ
0	[0.0, 0.0]	[0.0, 0.0]	[0.0, 0.0]	[0.0, 0.0]
1	[0.0]	[0.0]	[0.0]	[0.0]
2	[6.544413719711956e-14, 0.0]	[-7.470573096766964e-13, 0.0]	[0.0, 0.0]	[0.0, 0.0]
3	[]	[]	[]	[]
4	[0.0, -3.3223478222016434e-14]	[0.0, -7.547672881656753e-13]	[0.0, 0.0]	[0.0, 0.0]
5	[]	[]	[]	[]

	Muon/Muon.D0	Muon/Muon.DZ	Muon/Muon.ErrorD0	Muon/Muon.ErrorDZ
6	[]	[]	[]	[]
7	[]	[]	[]	[]
8	[]	[]	[]	[]
9	[]	[]	[]	[]
10	[]	[]	[]	[]
11	[0.0, 0.0]	[0.0, 0.0]	[0.0, 0.0]	[0.0, 0.0]
12	[0.0, 0.0]	[0.0, 0.0]	[0.0, 0.0]	[0.0, 0.0]
13	[]	[]	[]	[]
14	[]	[]	[]	[]
15	[]	[]	[]	[]
16	[0.0]	[0.0]	[0.0]	[0.0]
17	[-0.0, 0.0]	[0.0, 0.0]	[0.0, 0.0]	[0.0, 0.0]
18	[0.0, -0.0]	[0.0, 0.0]	[0.0, 0.0]	[0.0, 0.0]
19	[-0.0]	[0.0]	[0.0]	[0.0]
20	[]	[]	[]	[]
21	[0.0, -4.817514117662347e-14]	[0.0, -3.6103188581076973e-12]	[0.0, 0.0]	[0.0, 0.0]
22	[-0.0]	[0.0]	[0.0]	[0.0]
23	[-0.0, -2.962779794993351e-15]	[0.0, 4.9469292323547215e-14]	[0.0, 0.0]	[0.0, 0.0]
24	[]	[]	[]	[]
25	[0.0]	[0.0]	[0.0]	[0.0]
26	[0.0, -0.0]	[0.0, 0.0]	[0.0, 0.0]	[0.0, 0.0]
27	[0.0]	[0.0]	[0.0]	[0.0]
28	[0.0, -0.0]	[0.0, 0.0]	[0.0, 0.0]	[0.0, 0.0]
29	[0.0]	[0.0]	[0.0]	[0.0]

Esto se debe a que la targeta de delphes no contiene un módulo

```
module TrackCovariance TrackSmearing {
    (. . .)
}
```

Dado que solamente utiliza el módulo

```
module MomentumSmearing
```

El módulo *MomentumSmearing* se encarga de modificar el momento reconstruido, pero esto no genera una matriz de covarianza de los parámetros de la traza. En cambio, el módulo *TrackCovariance* es el encargado de asignar explícitamente *ErrorD0* y *ErrorDZ* a partir de dicha matriz.

II. Tarjetas de MadGraph

A. run_card.dat

Campo	V. Original	Prueba 01	Prueba 02
lpp1	1.0	1.0	1.0
lpp2	1.0	1.0	1.0
ebeam1	6500.0	6500.0	6800.0
ebeam2	6500.0	6500.0	6800.0
ptl	10.0	10.0	0.0
ptlmax	-1.0	-1.0	-1.0
etal	2.5	2.5	5.0
etalmin	0.0	0.0	0.0
drll	0.4	0.4	0.0
drllmax	-1.0	-1.0	-1.0
mmll	0.0	0.0	0.0
mmllmax	-1.0	70.0	10.0/70.0
mmnl	0.0	0.0	0.0
mmnlmax	-1.0	70.0	10.0/70.0
ptllmin	0.0	0.0	0.0
ptllmax	-1.0	-1.0	-1.0
xptl	0.0	0.0	0.0
ptl1min	0.0	0.0	0.0
ptl2min	0.0	0.0	0.0
ptl1max	-1.0	-1.0	-1.0
ptl2max	-1.0	-1.0	-1.0
maxjetflavor	4.0	4.0	4.0

B. delphes_card.dat

Apartado	**Set**	**Original y Prueba 01**
Muon efficiency ↓ *Módulo:* Efficiency MuonEfficiency	EfficiencyFormula{ }	(pt <= 10.0) * (0.00) + (abs(eta) <= 1.5) * (pt >10.0) * (0.95) + (abs(eta) >1.5 && abs(eta) <= 2.4) * (pt >10.0) * (0.95) + (abs(eta) >2.4) * (0.00)
Muon isolation ↓ *Módulo* Isolation MuonIsolation	DeltaRMax	0.50
	PTMin	0.50
	PTRatioMax	0.25

Apartado	**Set**	**Prueba 02**
Muon efficiency ↓ *Módulo:* Efficiency MuonEfficiency	EfficiencyFormula{ }	(pt <0.5) * (0.00) + (abs(eta) <= 0.1) * (pt >= 0.5) * (0.95) + (abs(eta) >0.1 && abs(eta) <= 5.0) * (pt >= 0.5) * (0.95) + (abs(eta) >5.0) * (0.00)
Muon isolation ↓ *Módulo* Isolation MuonIsolation	DeltaRMax	0.20
	PTMin	0.00
	PTRatioMax	0.25

Cambios a los módulos dentro de ExecutionPath{}

Original	**Prueba 03**
ParticlePropagator	ParticlePropagator
ChargedHadronTrackingEfficiency	ChargedHadronTrackingEfficiency
ElectronTrackingEfficiency	ElectronTrackingEfficiency
MuonTrackingEfficiency	MuonTrackingEfficiency
ChargedHadronMomentumSmearing	ChargedHadronMomentumSmearing
ElectronMomentumSmearing	ElectronMomentumSmearing
MuonMomentumSmearing	MuonMomentumSmearing
TrackMerger	TrackMerger
	ImpactParameterSmearing
ECal	ECal
HCal	HCal
Calorimeter	Calorimeter
EFlowMerger	EFlowMerger
EFlowFilter	EFlowFilter
PhotonEfficiency	PhotonEfficiency
PhotonIsolation	PhotonIsolation
ElectronFilter	ElectronFilter
ElectronEfficiency	ElectronEfficiency
ElectronIsolation	ElectronIsolation
ChargedHadronFilter	ChargedHadronFilter
	MuonFilter
MuonEfficiency	MuonEfficiency
MuonIsolation	MuonIsolation
MissingET	MissingET
NeutrinoFilter	NeutrinoFilter
GenJetFinder	GenJetFinder
GenMissingET	GenMissingET
FastJetFinder	FastJetFinder
FatJetFinder	FatJetFinder
JetEnergyScale	JetEnergyScale
JetFlavorAssociation	JetFlavorAssociation
BTagging	BTagging
TauTagging	TauTagging
UniqueObjectFinder	UniqueObjectFinder
ScalarHT	ScalarHT
TreeWriter	TreeWriter

Módulos

Original	# Track merger
	module Merger TrackMerger {
	add InputArrayChargedHadronMomentumSmearing/chargedHadrons
	add InputArray ElectronMomentumSmearing/electrons
	add InputArray MuonMomentumSmearing/muons
	set OutputArray tracks
	}
	# ECAL
	module SimpleCalorimeter Ecal {
	set ParticleInputArray ParticlePropagator/stableParticles
	set TrackInputArray TrackMerger/tracks
	set TowerOutputArray ecalTowers
	set EFlowTrackOutputArray eflowTracks
	set EFlowTrackOutputArray eflowPhotons
	(. . .)
	}

Prueba 03	# Track merger
	module Merger TrackMerger {
	add InputArrayChargedHadronMomentumSmearing/chargedHadrons
	add InputArray ElectronMomentumSmearing/electrons
	add InputArray MuonMomentumSmearing/muons
	set OutputArray tracks
	}
	# Track impact parameter smearing
	module ImpactParameterSmearing ImpactParameterSmearing {
	set InputArray TrackMerger/tracks
	set OutputArray tracks
	# Resolución de D0
	set ResolutionFormula {
	0.0116 + 0.0234/pt
	}
	}
	# ECAL
	module SimpleCalorimeter Ecal {
	set ParticleInputArray ParticlePropagator/stableParticles
	set TrackInputArray ImpactParameterSmearing/tracks
	set TowerOutputArray ecalTowers
	set EFlowTrackOutputArray eflowTracks
	set EFlowTrackOutputArray eflowPhotons
	(. . .)
	}